

NEXORA — SCIENTIFIC LUNAR MISSION REPORT

Target Region: Demo Surface (STANDARD)

RUN ID: TEST-01 | Location: Lat 0.00°, Lon 0.00° | Resolution: 1.0m Target

1. Selected Region & Geographic Provenance

LUNAR REGION NAME	Demo Surface	INPUT TMC-2 SPATIAL RES	5.0 meters
GEOGRAPHIC LAT / LON	Lat 0.00°, Lon 0.00°	TARGET GRID SPACING	1.0 meter (Derived)
DETECTED CRATERS	3 Craters	DETECTED BOULDERS	5 Boulders
SHADOW COVERAGE %	4.2%	TERRAIN CLASSIFICATION	Standard

2. Super-Resolution Reconstruction & Visual Imagery Panels

UPSCALING MODEL	LUNARSR	PSNR (RECONSTRUCTION)	28.5 dB	SSIM INDEX	0.88
-----------------	---------	-----------------------	---------	------------	------

3. Multi-Layer Hazard Map & Heatmap Analysis

FALSE-SAFE HAZARD RATE	3.1%	MEAN HAZARD SCORE	0.150	HAZARD MITIGATION	Significant vs 5m Raw
------------------------	------	-------------------	-------	-------------------	-----------------------

4. Selected Landing Zone & Explainable AI Diagnostics

CANDIDATE IDENTIFIER	Z-01	LANDING SUITABILITY SCORE	92.5 / 100
GRID COORDINATES [X, Y]	[250, 300]	SITE DECISION STATE	SAFE LANDING ZONE

Explainable AI Safety Diagnostics:

- [PASS] Safe slope clearance
- [PASS] Boulder threshold passed

5. Risk-Aware Lander Navigation Path

CALCULATED ROUTE DISTANCE	420.0 meters	MAX HAZARD ENCOUNTERED	0.15
AVERAGE ROUTE RISK	0.05	COMPUTATION LATENCY	120.0 ms

6. Quantitative Evaluation Benchmark Matrix

SUPER-RESOLUTION MODEL	PSNR (dB)	SSIM	HAZARD IOU	FALSE-SAFE %	PATH COST	LATENCY
TMC 5m (Raw Observed)	N/A	N/A	45%	18.0%	124.4	8ms
Bicubic 1m	23.1 dB	0.74	58%	11.8%	95.0	12ms
EDSR 1m	26.4 dB	0.83	75%	6.2%	77.9	185ms
SwinIR 1m	27.1 dB	0.85	79%	5.3%	73.4	420ms
LunarSR 1m (LUNARSR)	28.5 dB	0.88	86%	3.1%	65.5	120ms

SCIENTIFIC LIMITATION NOTICE: Super-resolved imagery is an estimated higher-resolution representation derived from lower-resolution source imagery. It must not be interpreted as direct physical observation. Slope, physical terrain roughness, local relief and elevation-dependent hazards require valid elevation data (DEM) or a separately validated terrain reconstruction method.